AKASHDIP MAHAPATRA
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PROFESSIONAL SUMMARY

Results-driven Systems Engineer with a strong foundation in mechanical **systems thinking**, currently managing **enterprise-scale AWS serverless architectures and CI/CD pipelines**. Proven ability to dive deep into **Linux fundamentals, network troubleshooting**, and cloud automation. Passionate about operational excellence, infrastructure reliability, and building highly available distributed systems.

CORE COMPETENCIES

- **Cloud Infrastructure:** AWS (EC2, S3, IAM, VPC, EventBridge, Lambda, MSK/Kafka, DynamoDB, ECR)
- **DevOps & CI/CD:** GitHub Actions, Docker, Jenkins, Terraform, Infrastructure as Code (IaC)
- **Operating Systems & Networking:** Linux (Ubuntu, RHEL), TCP/IP, DNS, HTTP/HTTPS, Nginx, Wireshark, nmap
- **Programming & Scripting:** Python (Boto3, Data Structures & Algorithms), Bash, C, SQL
- **Security & Observability:** Vulnerability Management (CVE patching), CloudWatch, Datadog, IAM Least Privilege

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | Onsite

*Data Engineer (DevOps & Infrastructure Focus under **British Airways**)* | Nov 2025 – Present

- **Manage** and **Automated** CI/CD deployment lifecycle (GitHub Actions to AWS ECR) for a multi-tiered AWS serverless data pipeline processing large-scale IoT and telemetry data.
- Support and troubleshoot a complex event-driven architecture utilizing Amazon MSK (Kafka), EventBridge, and containerized AWS Lambda functions to ensure high availability.
- Identify, investigate, and patch critical security vulnerabilities (CVEs) within Docker containers and Python dependencies, successfully reducing image sizes and hardening production security.
- **Engineered automated post-deployment validation workflows using Python, Boto3, and GitHub Actions, integrating CloudWatch and Datadog APIs to eliminate manual log-checking and accelerate deployment sign-offs.**

Tata Consultancy Services (TCS) | Remote

*Network & DevOps Trainee (**Apple Infrastructure** Project)* | Oct 2025

- Troubleshot deep Linux OS and networking issues, analyzing network traffic and performing advanced network forensics using tcpdump and Wireshark.
- Deployed and configured Nginx reverse proxies and load balancers, successfully resolving HTTP 403/404 errors and DNS routing failures.
- Hardened server security by implementing strict file permissions, writing custom Bash scripts for system automation, and configuring network firewalls (UFW, iptables, AWS Security Groups).

TECHNICAL PROJECTS

Custom Cloud Infrastructure & Automated CI/CD Pipelines

- Architected and deployed a highly secure custom VPN server utilizing an OpenVPN AMI on AWS EC2, managing traffic routing and network security boundaries.
- Built and maintained end-to-end CI/CD pipelines integrating GitHub Actions, Jenkins, and EC2 instances to automate the testing and deployment of software builds.

EDUCATION

Academy of Technology Engineering College | Hooghly, West Bengal

Bachelor of Technology in Mechanical Engineering | Sep 2021 – 2025

- Applied mechanical systems-design principles and analytical problem-solving skills to complex cloud infrastructure and software architecture.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Microsoft Certified: Azure Network Engineer Associate
- GitHub Certifications: Advanced Security, Actions, and Administration
- Oracle Cloud Infrastructure 2024 Certified Foundations Associate